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Jonathan Gruber; Maria Owings

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Physician financial incentives and cesarean section delivery

Jonathan Gruber*

and

Maria Owings**

The “induced-demand” model states that in the face of negative income shocks, physicians may exploit their agency relationship with patients by providing excessive care. We test this model using an exogenous change in the financial environment facing obstetrician/gynecologists: declining fertility in the United States. We argue that the 13.5% fall in fertility over the 1970–1982 period led ob/gyns to substitute from normal childbirth toward a more highly reimbursed alternative, cesarean delivery. Using a nationally representative microdata set for this period, we show that there is a strong correlation between within-state declines in fertility and within-state increases in cesarean utilization.

1. Introduction

■ Since the seminal work of Arrow (1963), it has been recognized that a central feature of the medical marketplace is the agency relationship between doctors and their patients. A standard model of this relationship in the health economics literature is that of “induced demand.” This model holds that in the face of negative income shocks, physicians may exploit this agency relationship by providing excessive care in order to maintain their incomes. If true, such a model has important implications for both the design of private-sector insurance policies and the optimal government response to rising medical costs. Despite its theoretical importance, however, there is only mixed empirical evidence on the role of inducement in physician decision making.

The inducement model has traditionally been tested by assessing how two alternative changes in the environment facing physicians affect the utilization of medical procedures. The first is reductions in the fees paid to physicians, generally by government payers. The second is variations in the physician/population density across areas; increased density lowers the income of the existing stock of physicians, and it will lead to increased utilization of medical procedures in an inducement-type model.

* Massachusetts Institute of Technology and National Bureau of Economic Research.

** National Center for Health Statistics.

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Each of these modelling approaches faces important problems, however. Fee changes cannot identify supply responses because there may be a contemporaneous demand response to changing prices; that is, if quantity rises when fees fall, this could reflect either induced or true increases in demand. Using fee changes to identify inducement also raises the problem that substitution and income effects go in opposite directions; even if there is no measured utilization response, there may be inducement, but substitution and income effects may simply be cancelling. Studies that use physician density differences to proxy for income shocks also face a fundamental identification problem. If there is some unobserved area characteristic that is correlated with taste for medical interventions, an area will feature both higher procedure utilization and more physicians, regardless of the extent of demand inducement. An obvious candidate for such an omitted variable is the average coinsurance rate in the area.

In this article, we propose a new means of identifying the effect of induced demand on procedure utilization. We exploit a plausibly exogenous change in the financial environment facing obstetrician/gynecologists during the 1970s: declining fertility in the United States. Fertility (births/100 population) fell by 13.5% during the 1970–1982 period. We argue that declining fertility increased the income pressure on ob/gyns and led them to substitute from normal childbirth toward a more highly reimbursed alternative: cesarean delivery. Used in only 5.5% of births in 1970, cesarean delivery rose by over 240% over the subsequent 12 years, and it is now the second most frequently performed major surgical procedure in the United States, with a rate of 23.5 cesarean deliveries per 100 births.¹ This time-series correlation between the fertility decline and the cesarean utilization increase raises the possibility that inducement played an important role in the substitution of cesarean delivery for normal childbirth.

Our primary goal is to test for a causal role of financial incentives in the use of cesarean delivery over the 1970–1982 period. We do so by exploiting the dramatic change in fertility patterns across U.S. states during this era. Conditional on the characteristics of the mother and of the birth, statewide fertility changes should provide exogenous measures of the financial pressure on ob/gyns in that state. Thus, we ask: In states where fertility was falling the most, did cesarean delivery rise the most?

Our test is carried out using a rich dataset of individual birth records from the 1970–1982 period. The National Hospital Discharge Survey (NHDS) is a nationally representative survey of hospitals that collects information for a sample of discharges in each hospital; it contains data on approximately 20,000 deliveries annually. We match these data on individual births to information on changes in fertility levels by state and year, and we examine how within-state changes in fertility affect the use of cesarean delivery. We use both individual-level and state-level data to control for a number of other characteristics of mothers and births, such as age, birth order, and infant health, which may be spuriously correlated with both the fertility rate and the use of cesarean delivery.

To summarize, we find a strong correlation between within-state fertility declines and within-state increases in cesarean delivery rates: a 10% decline in the fertility rate is associated with a .97 percentage point increase in the cesarean delivery rate. This finding is robust to a wide variety of specification checks. It implies that financial incentives did play an important role in the substitution of cesarean delivery for normal childbirth. This role was a small one relative to the rapid growth in cesarean delivery, however; increased ob/gyn income pressure can only explain between 16% and 32% of the substitution of cesarean delivery for normal childbirth during this era.

We begin, in Section 2, by discussing the causes and consequences of the diffusion of cesarean childbirth in the 1970s. In Section 3 we summarize the literature on the

¹ Unpublished tabulations, National Center for Health Statistics.

induced-demand hypothesis, and we present a brief model that notes the implications of this hypothesis for the use of cesarean delivery when the fertility rate falls. Section 4 describes the data sources and the empirical framework used. Section 5 presents the basic results and specification checks. Section 6 assesses whether cesarean delivery responded asymmetrically to fertility rises and falls. Section 7 concludes.

2. Background on cesarean delivery

■ **Cesarean delivery in the United States.** The upper line in Figure 1 charts the growth in the cesarean section rate (c-sections per 100 births) nationally over the 1970–1991 period. The rate doubled from 1970 to 1975, and doubled again by 1984. Utilization of the procedure flattened out in the mid-1980s and fell somewhat at the end of the decade. This rapid growth in c-section utilization was not restricted to any particular region, type of hospital, race, or age group (Placek and Taffel, 1980). The lower line in Figure 1 shows the growth in the cesarean section rate for women who had no previous c-sections (the primary cesarean rate); until recently, those with previous cesareans were almost universally c-sectioned on all future births.² The trend is similar, although a growing fraction of cesarean births over time is due to repeat cesareans.

There is no consensus view as to the cause of this rapid increase in cesarean utilization. Perhaps the most frequently cited explanation is the introduction of technologies for diagnosing fetal distress, such as electronic fetal monitoring. The percentage of deliveries with no reported complication fell from 70% in 1970 to 39% in 1984 (Shiono, McNellis, and Rhoads, 1987), and Williams and Hawes (1979) document a strong correlation between the percent of deliveries by cesarean section and the proportion of labor electronically monitored across California hospitals in 1977. However, cesarean section rates grew within diagnosis as well. In 1970, 11.6% of breech presentations were delivered by c-section; by 1984, 80% were delivered in this fashion (Shiono, McNellis, and Rhoads, 1987).³ National Institutes of Health (NIH) (1981) shows that almost one-half of the growth in primary cesarean rates was within-diagnosis growth.

Another important cause of c-section adoption was thought to be the legal environment. Marieskind (1979) reports that the most frequent reason given by physicians for performing a cesarean section was the threat of a malpractice suit. She cites a National Association of Insurance Commissioners study reporting that in 1975–1976, ob/gyns faced 10 times the probability of suit of other physicians. But recent econometric work that examines the regional correlation between malpractice “threat,” as measured by malpractice premiums or previous malpractice claims paid, and cesarean utilization has produced mixed results.⁴

□ **Financial incentives for cesarean delivery.** The hypothesis to be tested in the empirical work is that declining fertility led to a substitution toward cesarean delivery, in order to alleviate the income pressure on ob/gyns. A necessary precondition for this hypothesis is that cesareans be reimbursed more highly than normal childbirth, which has traditionally been true. The Health Insurance Association of America (1989) reports

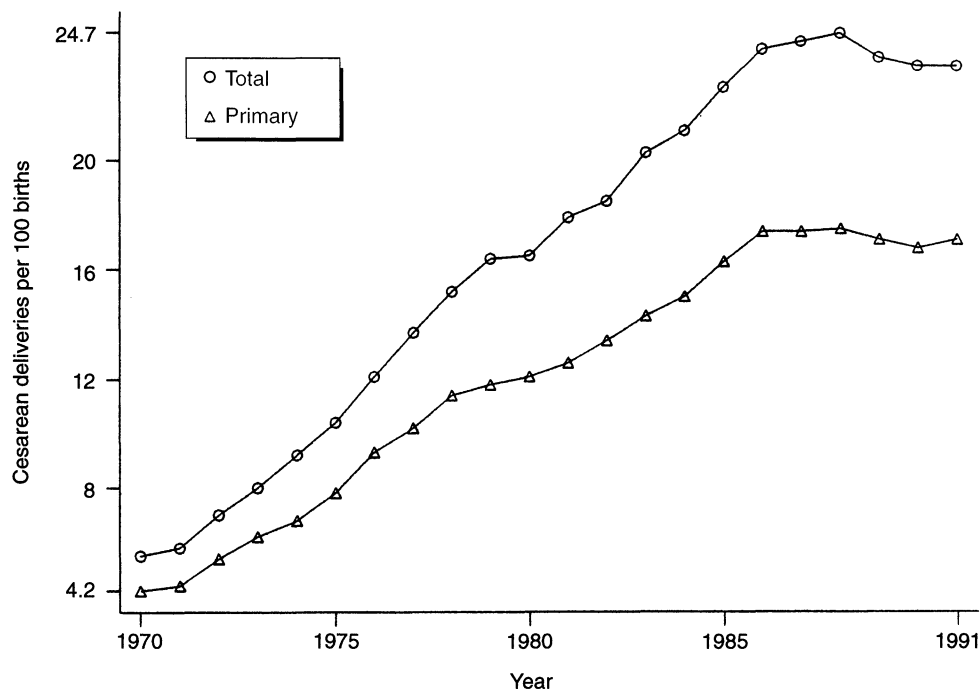
² In 1970, 98.3% of women with previous cesareans received c-sections; by 1984, the rate had fallen slightly to 96.1% (Shiono, McNellis, and Rhoads, 1987).

³ Breech presentations are births where the infant is not head facing when leaving the mother.

⁴ See Keeler and Brodie (1993) for a review, and Localio et al. (1993) and Tussing and Wojtowycz (1992) for recent results. Furthermore, none of these studies has been able to surmount the problem that there may be omitted regional variables correlated with both of the malpractice threat measures and cesarean utilization.

FIGURE 1

CESAREAN DELIVERIES IN THE UNITED STATES



that physician charges for cesarean delivery in 1989 were \$2,053, compared to \$1,492 for vaginal delivery. While no such systematic data are available for the 1970s, Marieskind (1979) reports fee differences for a sample of cities for the late 1970s: the average differential was approximately 40%, which is similar to the 1989 differential.

These higher reimbursement levels are not necessarily justified by increased physician inputs. In terms of just the time investment, cesarean delivery is much more efficient for the ob/gyn; unlike normal childbirth, cesarean deliveries can be scheduled in advance, and they often take less time than a normal childbirth with extended labor. Of course, cesarean delivery is a more difficult procedure. However, the 1993 revision of the resource-based relative value scale (RBRVS), which measures physician workloads by the product of intensity and time, concluded that the workload for vaginal delivery is actually higher than that of cesarean delivery (Keeler and Brodie, 1993).⁵

Furthermore, the structure of private insurance coverage during the 1970s may have contributed to the financial rewards of cesarean delivery. A commonly accepted feature of health insurance benefits before the mid-1970s was limited coverage for childbirth. Maternity coverage was sometimes excluded from basic health benefits; if included, it was often subject to flat rate cash amount limits, regardless of the cost of delivery. Many policies, however, limited coverage of normal childbirth only, and fully covered the costs of cesarean delivery. In this case, there would be an incentive for the physician to substitute cesarean childbirth for normal delivery. This differential coverage was outlawed by several state mandates in the 1975–1978 period and then by federal statute in October 1978 (Gruber, 1992).

⁵ This discussion does not account for any disutility from the extra training necessary for cesarean delivery; however, Marieskind (1979) notes that obstetricians were not even trained in vaginal breech delivery by the late 1970s, so that these training costs have presumably already been paid by practicing ob/gyns.

In theory, these state and federal laws changed the financial incentives of physicians in a testable way, by levelling the playing field between normal childbirth and cesarean delivery. This test runs into two problems in practice, however. First, while most private insurance did limit coverage of normal childbirth, many policies limited coverage of cesarean delivery as well. Gruber (1992) finds that 30% of private insurance plans in 1977 paid less than the physician's usual charges for childbirth than for other procedures. Thus, these laws, which typically introduced first-dollar coverage of a physician's usual charge for both normal and cesarean childbirth, simultaneously increased the incentives for both types of delivery. In addition, as noted below, we cannot identify privately insured patients in our data before 1977, which does not leave much variation in the legal environment before the federal statute of October 1978. Perhaps due to these problems, we were unable to find significant effects of these law changes on cesarean utilization. The inclusion of these variables, however, did not affect the coefficient on fertility in the regressions reported below.

There have been two types of previous studies of the role of financial incentives in the use of cesarean delivery. The first focuses on differences in cesarean utilization by insurance coverage type. The relative rewards for cesarean delivery for the privately insured are much higher than for the publicly insured or uninsured. Using data in American College of Obstetricians and Gynecologists (1989), we calculate that the average reimbursement differential for cesarean delivery in 1989 under Medicaid was only \$230, which is 40% as large as the average differential in physician charges discussed above. For uninsured patients, the financial rewards may be lower still, as there is a reasonable probability that the bill will not be paid at all. Epstein and Weissman (1989) find for a sample of Massachusetts hospitals that 80% of total charges to self-pay patients are not paid. While similar figures for physicians' charges are not available, this suggests that physicians may be unable to recoup from uninsured patients the differential charge for doing a cesarean delivery.

Stafford (1990) examined cesarean rates by payment source in a sample of California hospitals and found results that lined up quite well with these financial incentives: cesarean rates were highest for those with private insurance, followed by the publicly insured, and then the uninsured. It is difficult, however, to draw causal inferences from this fact because there may be omitted characteristics by payer class that are correlated with the cesarean decision (i.e., birth severity).

The second test focuses on the supposition, made for example by Marieskind (1979), that a declining fertility rate and increases in ob/gyn density may have led to increased use of cesarean delivery in the 1970s; this point is developed theoretically below. Tussing and Wojtowycz (1992), examining New York data, find no evidence that cesarean delivery rates are higher where there is a higher ob/gyn density. Once again, however, such a study suffers from the fact that there may be omitted regional differences that are correlated with both higher ob/gyn density and cesarean utilization.

□ **Consequences of c-section diffusion.** One potential cause of the diffusion of cesarean sections was the belief that they were instrumental in improving birth outcomes. There is a strong time-series correlation between the rapid improvement in birth outcome in the United States and the diffusion of cesarean sections. From 1968 to 1983, the infant mortality rate in the United States was halved; this is the exact period that saw rapid diffusion of cesarean sections. However, Marieskind (1979) is critical of this interpretation of the time-series data, noting that there were many other technological advances in the treatment of childbirth, such as the development of neonatal intensive care units.

Microdata evidence on the effectiveness of cesarean delivery is mixed. Williams and Hawes (1979) studied a sample of hospitals in California in 1977 and found a

negative correlation between increased use of technology in childbirth, including fetal monitoring and cesarean delivery, and the demographically adjusted perinatal mortality rate. On the other hand, in a sample of mature (over 2,500 grams) breech births in New York City, the NIH (1981) found no change in survival probabilities over time, despite a sizable rise in the cesarean delivery rate. And Finkler and Wirtschafter (1993) find no relationship between cesarean section use and risk-adjusted birth outcomes across their sample of eight California hospitals.

While we are unaware of any randomized clinical trials to evaluate the effectiveness of cesarean sections, there have been several trials that randomly assigned the availability of electronic fetal monitoring. Three of the four studies found that monitoring was associated with higher c-section rates but no improvements in perinatal outcomes (NIH, 1981).

Cesarean delivery is also more costly than vaginal childbirth, along two dimensions. The first is physiological costs. Maternal mortality is very low overall (approximately one death per 10,000 births), but it is two to four times as high for cesarean as for vaginal deliveries (NIH, 1981).⁶ There are a number of less severe but more common risks as well, such as postoperative infection and hemorrhage. There are also potential physiological costs to the infant, such as respiratory distress syndrome in infants who are delivered prematurely by c-section (Marieskind, 1979).

The second is financial costs. In 1989, the total (doctor and hospital) costs of a cesarean delivery were 66% higher than the costs of a normal delivery (Health Insurance Association of America, 1989). In that year, there were 938,000 cesarean section deliveries in the United States (Graves and Kozack, 1992). If these deliveries had been as costly as the average normal childbirth, then this would account for a differential of \$2.7 billion dollars in medical spending. While this is clearly an upper bound, since cesarean deliveries are likely to be more severe, it does highlight the large differential expenditure on cesarean delivery in the United States.

3. Induced demand: theory and evidence

■ **An inducement model.** The induced-demand hypothesis has been explored by a number of studies over the past 20 years. The building block of this theory is the notion that physicians, as agents of uninformed patients, are able to determine the level of demand for their product.⁷ Physicians are generally posited to have utility functions that depend positively on their income, leisure, and on delivering the “appropriate” amount of care. The last factor can enter either through the physician’s “internal conscience” (McGuire and Pauly, 1991) or as a result of a reputation process by which physicians who excessively induce demand are punished through future reductions in true patient demand (Dranove, 1988).⁸ Inducement can be defined as the deviation of treatment levels from that which equates the marginal benefits and costs to patients (Fuchs, 1978).

A natural implication of inducement models is that a fall in the birth rate will lead to adoption of more highly reimbursed procedures such as cesarean delivery. This is readily illustrated in a simple inducement model along the lines of McGuire and Pauly (1991).⁹ We take the number of births as given to the ob/gyn.¹⁰ We also assume that

⁶ More recent data on maternal mortality may show a smaller differential between vaginal and cesarean deliveries.

⁷ Dranove (1988) correctly notes that induced-demand models can be equally well applied to other agency relationships between buyers and (expert) sellers.

⁸ For earlier models of this type of provider behavior, see Evans (1974) and Newhouse (1970).

⁹ We are grateful to Tom McGuire for suggesting this formulation of the McGuire-Pauly model for our case.

¹⁰ This may be particularly inappropriate in the 1970s, since most insurance plans did not cover the cost of normal childbirth and so demand may have been somewhat price elastic. However, even if the number of births for a given ob/gyn is price elastic, changes in the stock of births in the state still provide an exogenous shock to the income of the average ob/gyn in that state.

the only margin for inducement is to increase intensity by shifting the nature of the delivery from a normal childbirth to a cesarean delivery. This is not strictly true, since ob/gyns appear to earn only about half of their income from obstetrics on average.¹¹ Introducing other dimensions of inducement into the model would not change the direction of our prediction, however; whether inducement in fact takes place through the type of delivery is ultimately an empirical question.

We posit that the ob/gyn has a utility function of the form

$$U = U(Y, I), \quad (1)$$

where Y is full income, earnings minus the value of forgone leisure, and I is the extent of "inducement," which causes the physician disutility through either the "internal conscience" or "demand reduction" channels ($U_Y > 0$, $U_I < 0$; U_{YY} , $U_{II} < 0$). The number of births is B , and the level of inducement per birth is i , so that $I = Bi$; the total disutility from inducement depends on the total amount of inducement across all patients. The fraction of cesarean deliveries is $a(i)$, $a' > 0$, $a'' = 0$. Thus, the budget constraint is

$$\begin{aligned} Y &= Y_n N + Y_c C, \\ C &= Ba(i), \\ N &= B(1 - a(i)), \end{aligned} \quad (2)$$

where N is the number of normal births and C is the number of cesareans. Y_n and Y_c are the full incomes from performing normal and cesarean deliveries, respectively. We assume that $Y_c > Y_n$. That is, we assume that the reimbursement premium for cesarean deliveries is sufficiently high to compensate the physician for any loss in leisure from performing a cesarean. This seems reasonable given the data presented above that cesarean fees are much higher, that cesarean delivery takes less time, and that the increased difficulty of cesarean delivery does not, in the judgment of the experts who developed the RBRVS, compensate for the time reduction.

We follow McGuire and Pauly (1991) in assuming separable utility. The first-order condition becomes

$$U_Y a' [Y_c - Y_n] + U_I = 0. \quad (3)$$

That is, the physician trades off the net disutility from inducement against the net utility from increasing income through shifting to cesarean delivery. Suppose now that the number of births, B , is exogenously reduced. Fully differentiating (3), we obtain

$$\frac{\delta i}{\delta B} = \frac{-U_{YY}(Y/B)a'(Y_c - Y_n)}{U_{II}} < 0. \quad (4)$$

That is, as births fall, inducement per birth rises, so that the rate of cesarean delivery rises. We discuss the magnitude of this incentive in more detail in Section 4.

¹¹ American College of Obstetricians and Gynecologists (1992) reports that 56% of ob/gyns earn at least 40% of their incomes, and 76% earn at least 25% of their incomes, from obstetrics. Furthermore, data from this source indicate that 55.8% of ob/gyn visits are for prenatal or postnatal care, and American Medical Association (1993) reports that one-half of all surgical procedures performed by ob/gyns are childbirth.

□ **Physician type, mobility, and entry.** Two important further assumptions underlie this model. First, we have assumed that all births are delivered by ob/gyns. If in fact a nontrivial share of normal births is delivered by practitioners other than ob/gyns, then ob/gyns may be insulated from changes in the birth rate. That is, deliveries by non-ob/gyns may fully absorb fluctuations in the number of births, so that ob/gyns have no change in inducement incentives. If only ob/gyns perform cesarean deliveries, then this scenario could in fact lead to a spurious (from our perspective) negative correlation between fertility changes and cesarean utilization.

The vast majority of births, however, are delivered by ob/gyns. During our sample period, over 99% of all births were delivered by a physician in a hospital setting (U.S. DHHS). Of physicians, the American Medical Association (AMA) (1993) reports that ob/gyns perform 13.8 deliveries per month, while general and family practitioners (the second-largest category) perform only .7. The number of general and family practitioners is approximately twice that of ob/gyns, so ob/gyns deliver over 90% of all babies by this calculation. Similarly, using data on the fraction of dollars derived from delivery services for a large Blue Cross/Blue Shield plan in a major U.S. city, we tabulate that 87% of delivery dollars were attributable to ob/gyns.

Second, we have assumed that ob/gyns are immobile. If ob/gyns are mobile, then their location may respond to fertility changes in a way that partially offsets the inducement incentives documented above. In fact, it does appear that ob/gyns respond to changes in fertility rates in making their locational decisions; a regression over the 1970–1982 period of the log of the number of ob/gyns on the log of the fertility rate, including a full set of state and year dummies, yields a significant and positive coefficient of roughly .2. To the extent that ob/gyns are offsetting fertility declines by their locational choices, it will lead to a smaller effect of fertility changes on the use of cesarean delivery in the empirical work below. Whether these locational choices are sufficient to fully undo the need for inducement in response to falling fertility is an empirical question.

There is also a simple fact that seems to contradict the contentions of this model: entry into obstetrics and gynecology continued unabated during the era of fertility decline, with ob/gyns per capita increasing by 38% from 1970 to 1982. If fertility decreases were increasing the financial pressure on ob/gyns, why should they have continued to enter this field at such a rapid rate? There are a number of other factors, however, that may have affected the attractiveness of medical professions during this era, such as the general decline in returns to higher education of the 1970s (Freeman, 1976). A more relevant question is whether entry into this field relative to other medical subspecialties responded to fertility changes. In fact, there is a strong positive correlation (.75) between annual fertility rates and the ratio of ob/gyns to other physicians in the United States over the 1970–1982 period. This suggests that, conditional on choosing to enter the medical profession, individuals were considering the relative rewards of this field in making their specialty decision.

□ **Induced demand: past evidence.** As noted above, induced-demand effects have been posed and tested in two different contexts. The first is reductions in the fees paid to physicians, generally by government payers. In this case, inducement operates through both the substitution and income effects, so that the net effect on the quantity of services is unclear. However, most previous studies have tested for inducement by assessing whether quantity increased when fees were reduced (that is, under the assumption that income effects dominate substitution effects); in the limit, with a “target income,” quantity increases fully undo the price reductions.

There is some evidence that fee reductions lead to increases in quantities, although the magnitude of inducement is very uncertain. The strongest evidence for induced-quantity increases is Rice's (1984) study of a Medicare fee reduction in Colorado; he found that a 10% decline in physician reimbursements led to a 6.1% increase in intensity of medical services and a 2.7% increase in intensity of surgical services. However, a similar study by Hurley, Labelle, and Rice (1990) for Ontario found mixed responses to fee changes across procedures. Evidence from the experiences of the United States and Canada with fee controls offers a similarly mixed picture of the role of induced demand; see Wedig, Mitchell, and Cromwell (1989), Feldman (1988), Feldman and Sloan (1989), and Rice and Labelle (1989) for a discussion.

The second context for testing induced demand is the response of procedure use to variations in the physician/population density across areas. Evidence on the use of surgery in response to an increased supply of surgeons is presented in Fuchs (1978) and Cromwell and Mitchell (1986). Both authors looked at a cross-section of locations over several years and found strong evidence for demand inducement; the latter findings are about one-third as large as the former. Birch (1988) and Grytten, Holst, and Laake (1990) also find very strong positive relationships between dentists per capita and utilization of dental services.¹²

As discussed in the Introduction, both of these methodologies face important identification problems. The problem of omitted variables correlated with physician density is discussed in Auster and Oaxaca (1981), Phelps (1986), and Feldman and Sloan (1989). Both Fuchs and Cromwell and Mitchell attempt to surmount this problem by finding instruments correlated with physician density, such as population distribution across metro and nonmetro areas, average wages, and other socioeconomic characteristics of the area. It is unclear, however, whether these instruments are also uncorrelated with taste for surgery; wealthier or more urban areas may feature different average coinsurance rates.¹³

In this article, we propose an alternative means of identifying an inducement effect for ob/gyns: using changes in the fertility rate. The model above shows that changes in fertility will have inducement-type incentives. Moreover, conditional on the characteristics of the mother and the birth, statewide changes in fertility should be an exogenous shock to the income of ob/gyns. In the empirical work below, we shall consider this exogeneity assumption in more detail, by controlling for a number of potential omitted correlates of fertility rate changes and cesarean delivery rates.

In some specifications below, we shall also follow the previous literature in using changes in ob/gyn density to proxy for income pressure. We potentially improve on previous estimates of the effect of physician density on procedure utilization by including location (state) fixed effects in the regression, thus controlling for any omitted variables that are constant over time. Nevertheless, some of the identification problems noted by the critics of this approach are difficult to surmount simply by using within-area changes. Thus, we shall primarily rely on fertility changes to proxy income pressure on ob/gyns in the estimation.

4. Data and empirical framework

■ **Data.** Our primary data source is the National Hospital Discharge Survey (NHDS) conducted annually since 1965 by the National Center of Health Statistics (NCHS).

¹² For reviews of the induced-demand literature, see Feldman and Sloan (1989) or Dranove and Wehner (1994).

¹³ This criticism is levelled most damningly by Dranove and Wehner (1994), who find, using an instrumental variables methodology similar to that of Cromwell and Mitchell (1986), that increasing the number of ob/gyns in an area increases the number of births.

This is a nationally representative survey of over 400 hospitals, and it contains information on approximately 200,000 discharges in each year. The survey collects information on some limited demographic characteristics of the patient, features of the hospital, primary and secondary diagnoses, and primary and secondary procedure use. We use information on patients with a primary diagnosis of childbirth in the years 1970–1982; the final sample consists of over 250,000 births. The public use file of the NHDS identifies only the region of the hospital. These files were augmented with state identifiers, and all analysis was carried out at the NCHS.¹⁴

The means for the dataset are presented in Table 1, both for the full dataset and for births that did and did not get delivered by c-section.¹⁵ Approximately 11.9% of births are by cesarean section over this period. Although the majority of births are to women under age 25, the majority of c-sections are performed on women over age 25. There are relatively few deliveries in for-profit hospitals in this period. The most striking differences between the cesarean and noncesarean columns are in the figures for hospital size and for severity codes; cesareans are much more likely in large hospitals, and they are strongly correlated with a complicated delivery. Since our identifying variation will come from state-level changes in fertility, it is comforting to note that we have a reasonably large average sample size of deliveries for each state per year. The mean number of deliveries by state per year is 837, and the value ranges from a minimum of 17 to a maximum of 2,767.

To examine the effect of fertility on the use of cesarean delivery, we exploit the substantial variation across states in fertility changes in this era. Data on the number of births in each state and year are available from the NCHS publication *Vital Statistics of the United States* (U.S. DHHS).¹⁶ Figure 2 graphs the average fertility rate, births per 100 population, by year for our sample period. There is a striking decline in fertility from 1970 to 1975, and a moderate rise from 1977 to 1982. Over the entire period, fertility falls from 1.84 births per 100 persons to 1.59 births per 100 persons, a decline of 13.5%.¹⁷ These national trends mask substantial heterogeneity in fertility changes within states over time. Figures 3 and 4 illustrate two examples of this heterogeneity. In both North Carolina and Kansas, there were steep fertility declines in the early years of the sample period. But while in North Carolina fertility declined until 1976 and was flat thereafter, in Kansas fertility dropped much more rapidly and rose thereafter, so that by 1982 fertility levels were actually higher than in 1970. Ranking states by their fertility rates in 1970 and again in 1982, there is only a .58 correlation in the ranking across the two years, and 28 states changed their ranking by more than 10 positions during this period.

What drove these differential fertility changes across the states? There are a number of hypotheses for the nationwide changes in fertility during this period: postponement of marriage and rising divorce rates; increasing use of reliable contraceptives; liberalized laws

¹⁴ The NHDS does not collect data for three low-population states, so these states were dropped from the analysis. The NHDS also provides weights to capture the probability that a given discharge was sampled. All results were computed both weighted and unweighted, and were very similar in every case. The results reported below are unweighted.

¹⁵ The NHDS identifies expected source of payment only from 1977 onward, so we do not use that variable in our basic analysis. We do use it in one specification check below.

¹⁶ Ideally, the fertility rate would be calculated as the ratio of births to the female population of child-bearing age. However, population by state/sex/age group is not available between 1970 and 1980, so we use total population in our basic estimates. Attempting to correct our regressor for trends in the demographic composition of each state's population has little effect on the results.

¹⁷ Fertility fell dramatically as well from 1960–1965 and then was roughly constant until 1970. Data on cesarean delivery are not available for this earlier period. Our assumption is that cesarean utilization had adjusted to the earlier transition by 1970, so that we can assess the net new contribution of the change from 1970 onward.

TABLE 1 Means of the Dataset

	All Births	No C-Section	C-Section
Cesarean delivery	.119	0	1
Age <20	.177	.183	.134
Age 20–25	.341	.346	.308
Age 25–30	.296	.293	.321
Age 30–35	.132	.128	.165
Age 35–40	.041	.039	.055
Age 40–45	.010	.009	.014
Age 45+	.002	.002	.003
Married	.793	.792	.798
White	.693	.692	.698
Black	.163	.163	.167
Number of beds	360.5 [301.2]	356.8 [302.5]	388.2 [389.4]
For-profit	.035	.035	.041
Government	.261	.264	.237
Previous cesarean	.035	.001	.286
Breech presentation	.028	.017	.107
Fetal distress	.045	.015	.272
Maternal distress	.022	.014	.083
Other complications	.329	.305	.509
Ob/gyns per 100 births	.621 [.248]	.614 [.246]	.671 [.261]
Ob/gyns per 100 population ($\times 10^2$)	.954 [.304]	.946 [.301]	1.020 [.319]
Births per 100 population	1.59 [.23]	1.60 [.24]	1.57 [.23]
Number of observations	255,593	225,202	30,391

Notes: Data from the NHDS dataset and other sources are described in the text. Standard deviations are in square brackets.

on abortion;¹⁸ concern about environmental issues and overpopulation; increased entry into and opportunities in the labor force for women; and cyclical conditions.¹⁹ Both Gibson (1976) and Bloom and Trussell (1984) find that this national fertility decline was composed of both permanent fertility effects and changes in the timing of births; some of the “bounceback” in fertility in the later part of the decade may have been a result of these timing effects. We are aware of no state-specific analyses for these years, however.

This article will not explore the determinants of these cross-state fertility changes; rather, we shall take them as exogenous local shocks to the income of ob/gyns that can be used to test the induced-demand model. A number of alternative hypotheses can be suggested for a fertility/cesarean correlation, and we address these at length in Section 5. While it is difficult to convincingly refute all possible alternative hypotheses for this correlation, we find it to be remarkably robust to any alternative considered.

¹⁸ Westoff and Ryder (1977) document a large decline in unwanted births during the 1960s and conjecture that this continued into the 1970s. Statistics Canada (1984), in analyzing parallel demographic trends in Canada, noted that in 1968, 16% of couples in Quebec did not want their last child; by 1976, this ratio had declined to 7%.

¹⁹ Butz and Ward (1979) document countercyclical fertility, which they explain as a function of fertility responding to changing labor market opportunities for women. Although there was a recession during the tail end of this period of fertility decline, the economy was growing at its start.

FIGURE 2

FERTILITY OVER TIME

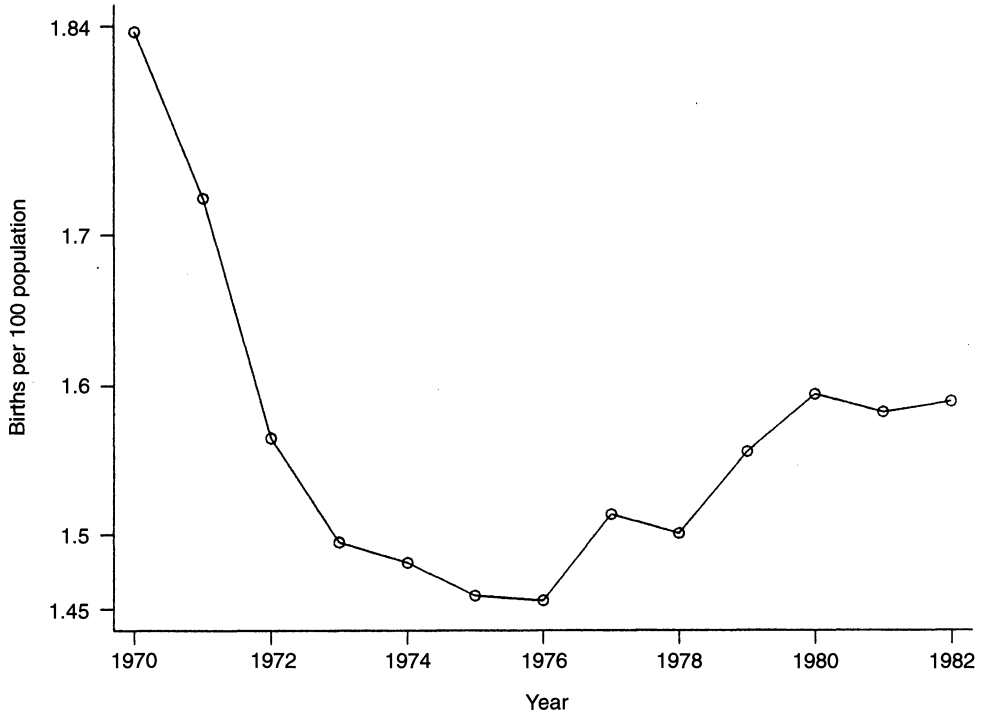


FIGURE 3

FERTILITY OVER TIME IN NORTH CAROLINA

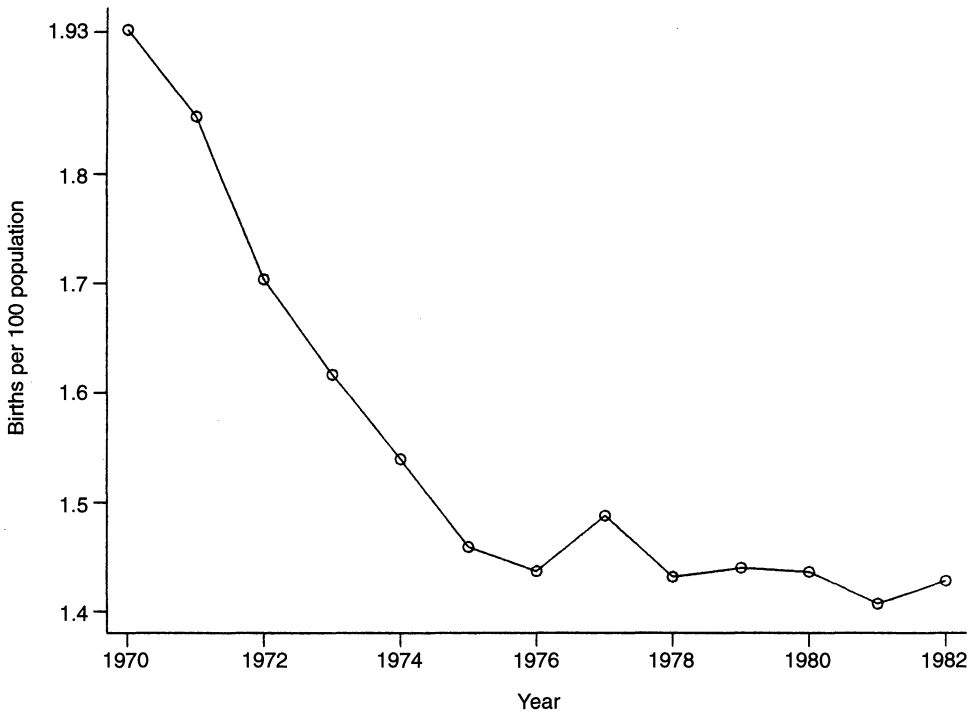


FIGURE 4
FERTILITY OVER TIME IN KANSAS



How large is the financial incentive arising from these differential fertility changes? We noted above that half of ob/gyn income is from obstetrics, and fertility fell by 13.5% during the 1970–1982 period. Assuming that, in the absence of a behavioral response, the decline in income from obstetrics would parallel the decline in the birth rate, the implied fall in ob/gyn income is 6.75%, which is nontrivial. If the trend in ob/gyn density is taken to be exogenous, then the pressure on ob/gyn income is even greater: ob/gyns per birth rose by 58% in this era. Moreover, ob/gyn density increases could affect both income from obstetrics and gynecology, so this represents a negative income shock on the order of 30% or more.

As Table 1 shows, the means for ob/gyn density and fertility support our hypothesis of induced demand: cesareans are more likely where there are more obstetricians and fewer births.²⁰ However, this hypothesis is most appropriately tested in a multivariate regression framework, which we describe next.

□ **Empirical framework.** To assess the effect of financial incentives on the cesarean decision, we initially run logit regressions of the form

$$CSEC_{iht} = f(\alpha + \beta_1 X_{iht} + \beta_2 PREV_{iht} + \beta_3 Z_{ht} + \beta_4 LOBKID_{jt} + \beta_5 \delta_j + \beta_6 \tau_t + \epsilon_{iht}), \quad (5)$$

where

²⁰ Data on ob/gyn density in each state and year is from AMA (1993).

$CSEC_{ihjt}$ is equal to one if individual i in hospital h in state j in year t received a c-section, and zero otherwise,

X is a set of individual demographic characteristics,

$PREV$ is an indicator for previous cesarean delivery,

Z is a set of hospital characteristics,

$LOBKID$ is the log of the number of ob/gyns per birth,

δ_j is a full set of state dummy variables, and

τ_t is a full set of year dummy variables.

In this regression, the cesarean decision is modelled as a function of a number of characteristics of the mother and the hospital, as well as the ob/gyn density (ob/gyns per birth) in the state. If ob/gyn locational choice can be taken as exogenous, then the ob/gyn density provides the appropriate measure of the negative income shock for ob/gyns; cesareans are more likely to be induced where either the supply of ob/gyns is rising or the birth rate is falling. However, as documented above, we are concerned that the number of ob/gyns does not provide an exogenous measure of financial pressure, so our preferred specification is

$$CSEC_{ihjt} = f(\alpha + \beta_1 X_{ihjt} + \beta_2 PREV + \beta_3 Z_{hjt} + \beta_4 LFERT_{jt} + \beta_5 \delta_j + \beta_6 \tau_t + \epsilon_{ihjt}), \quad (6)$$

where $LFERT$ is the log of the state/year fertility rate (births per 100 population). In this specification, we identify the effect of financial pressure solely from changes in fertility. We normalize births by population because, since we are no longer explicitly including the number of ob/gyns in the state, population growth provides an exogenous indicator of the trend growth in ob/gyns (and thus the underlying growth in births needed to maintain ob/gyn income). Our conclusions are not sensitive to the use of this normalization. Table A1 in the Appendix presents the basic regression results using log births, log births along with log population, log fertility, and the level of fertility. In each case, the coefficient on the regressor of interest is highly significant in the expected direction, although the magnitudes of the effects vary somewhat across the columns. When log birth and log population are entered separately, they have almost identical and opposite (statistically significant) coefficients, supporting the contention that a normalization by population is appropriate.

We control for the age of the mother with a series of dummy variables, since the likelihood of cesarean delivery rises sharply with age; we include dummies for five-year age intervals from 20 to 45, and for 45 and above (the omitted group being those under age 20). We also include indicator variables for whether the mother is white or black (other nonwhite being the excluded group) and an indicator for married mothers. We control for whether the mother is coded as having had a previous cesarean, since cesarean delivery is almost certain in this case.²¹ At the hospital level, we control for hospital bed size, and whether the hospital is for-profit or government owned (nonprofit being the excluded group).²² We include a full set of year dummies in all regressions,

²¹ In our sample, 97% of women with previous cesarean childbirth had their births by cesarean delivery. The previous indicator may also be capturing some effects of birth order, since previous is zero by definition for the first birth, and first births are more likely to be delivered by cesarean section (Marieskind, 1979).

²² Unfortunately, data on teaching status of the hospital are not available, although there is likely to be little within-state change in teaching status over this period that is not captured by changes in hospital size or ownership.

since there was a strong trend in cesarean delivery over this period. Finally, we include a full set of state dummies so that we can control for fixed differences across areas that might be correlated with both fertility and the cesarean decision.

5. Results

■ **Basic results.** Table 2 presents the basic regression results. The first column reports the results from running equation (5). There is a significant positive relationship between the log of ob/gyn density and the probability of cesarean delivery: a 10% increase in the ob/gyn density raises the probability of a cesarean by .6 percentage points.²³

The likelihood of cesarean delivery rises with age, as indicated by Table 1. The increase is monotonic until the highest age group, where there is some reduction; however, this category contains only a very small number of observations, and an overall increase with age cannot be rejected. Cesarean delivery is more likely for married mothers and white mothers. Previous cesarean delivery, as expected, is an incredibly strong predictor of current cesarean delivery. Cesarean delivery is much more likely in large hospitals. Cesareans are also somewhat more likely at for-profit hospitals, and much less likely at government hospitals, than at nonprofit hospitals. The state and year dummies (not reported) are highly jointly significant.

In column (2), we disaggregate the doctor/birth ratio into the log of ob/gyns per 100 population and the log of the fertility rate. Both coefficients are supportive of the induced-demand view; that is, cesarean rates were rising both where the number of ob/gyns was increasing and where the birth rate was declining. Both coefficients are statistically significant. In column (3) we exclude the ob/gyn ratio altogether. The effect of fertility is somewhat strengthened; a fall in the fertility rate of 10% is associated with an increase in the likelihood of cesarean delivery of .97 percentage points. In column (4), we exclude the fertility rate and examine the effect of changes in ob/gyns per 100 population. The effect of ob/gyn density increases somewhat.

How large are these effects? A 10% decline in the fertility rate approximately translates to a 5% drop in the income of ob/gyns, if they do not change their behavior. In 1989, cesarean delivery was reimbursed \$561 more highly than normal childbirth, and the typical ob/gyn performed 168 births per year (AMA, 1993). A one percentage point rise in cesarean utilization translates to 1.68 more cesarean deliveries per year, or an income increase of \$943. In 1989, the average ob/gyn income was \$194,300 (*Medical Economics*, September 3, 1990), so this represents an income rise of only .5%. This suggests that the shift to cesarean delivery offsets only a very small part of the negative income shock facing ob/gyns.²⁴

An alternative means of scaling the result is to calculate the implied contribution of the 1970–1982 fertility decline to the growth in cesarean utilization over this period. Viewing ob/gyn density as exogenous, our finding in column (1) implies that the increase in ob/gyn density over the 1970–1980 period can explain 2.9 percentage points of the 9.1 percentage point increase in primary cesarean delivery rates (32%).²⁵ Alternatively, viewing only the fertility differences as exogenous, we find that the fall in

²³ The logit coefficients are not directly interpretable, but they can be roughly translated to probabilities by multiplying the coefficients by $p/(1-p)$, where p is the sample mean cesarean section rate (.119).

²⁴ Unfortunately, data do not exist to permit a comparable calculation for the 1970s, but the inference is undoubtedly quite similar in that era, given that Marieskind (1979) reports similar cesarean delivery fee premiums for the late 1970s (as noted above).

²⁵ In this calculation and the ones to follow, we compare the estimated effect of our inducement indicators to the growth in primary cesarean deliveries, not total cesarean deliveries. We do this because, by conditioning on previous cesarean delivery, we are modelling the effect of inducement on the primary cesarean decision. In fact, if these models are estimated only on the sample without previous cesarean delivery, the results are almost identical, as would be expected given the very small likelihood of vaginal birth after cesarean.

TABLE 2 **Basic Regression Results**
Dependent Variable is a Dummy for Cesarean Delivery

	(1)	(2)	(3)	(4)
Log ob/gyn per 100 births	.578 (.126)			
Log fertility (births per 100 population)		-.888 (.202)	-.923 (.201)	
Log ob/gyn per 100 population		.342 (.174)		.410 (.174)
Age 20–25	.001 (.022)	.001 (.022)	.001 (.022)	.001 (.022)
Age 25–30	.081 (.023)	.081 (.023)	.082 (.023)	.082 (.023)
Age 30–35	.153 (.027)	.153 (.027)	.153 (.027)	.154 (.027)
Age 35–40	.419 (.037)	.418 (.037)	.418 (.037)	.419 (.037)
Age 40–45	.616 (.065)	.616 (.065)	.616 (.064)	.615 (.064)
Age 45+	.341 (.143)	.341 (.143)	.342 (.143)	.342 (.143)
Married	.018 (.020)	.019 (.020)	.019 (.020)	.018 (.020)
White	.025 (.022)	.026 (.022)	.026 (.022)	.024 (.022)
Black	-.003 (.030)	-.003 (.030)	-.003 (.030)	-.004 (.030)
Previous cesarean	5.760 (.064)	5.761 (.064)	5.761 (.064)	5.760 (.064)
Number of beds ($\div 10^3$)	.263 (.025)	.264 (.025)	.263 (.025)	.264 (.025)
For-profit	.091 (.039)	.092 (.039)	.094 (.039)	.093 (.039)
Government	-.197 (.020)	-.198 (.020)	-.198 (.020)	-.197 (.020)

Notes: Standard errors are in parentheses. Regressions are run as logits. All regressions include a full set of state and year dummies. $N = 255,593$.

fertility in the United States from 1970 to 1982 can account for 1.45 percentage points of primary cesarean increase, which is only 16% of the growth rate of cesarean delivery over this period.

To summarize, our basic findings are supportive of the hypothesis that ob/gyns responded to falling fertility in this era by performing more cesarean section deliveries. The estimated response is small, however, relative to the shock to ob/gyn income, and our finding implies that the general fertility decrease during this era can explain only a small part of the overall growth in cesarean utilization.

□ **Specification checks.** We find the results in Table 2 to be striking evidence for inducement. It is important to note, however, that there are a number of alternative interpretations of this finding than the one we have given. We are unable to refute all of these alternatives. In this section, we consider several of the most natural ones.

The first is that the average riskiness of births increases as the number of births declines. In this case, the use of cesarean delivery may change, due not to physician

financial incentives but to the natural correlation between high-risk births and the use of c-sections. For example, suppose that there are two types of mothers: those who are both careful with contraception and take good care of their fetuses ("prudent" mothers), and those who do neither ("imprudent" mothers). Fertility among imprudent mothers in this model will be roughly constant, so that, as fertility drops, a higher fraction of births will be represented by imprudent mothers. This will lead to a positive correlation between the fertility rate and the quality of the fetus, and thereby a spurious (from our perspective) negative correlation between the fertility rate and the cesarean section rate. Microdata studies have shown that very low or high birthweight, or very short or long gestation, are correlated with the use of cesarean delivery (Marieskind, 1979).

We have already controlled for some characteristics of the mother (i.e., age) that might indicate a high-risk birth. In Table 3, we consider two further steps; in each case, we report only the coefficients of interest, as the other covariates see little change in their marginal effects. In the first column, we control for three indicators of the underlying riskiness of births in state j in year t : the fraction of live births that die within 28 days (the neonatal mortality rate), the fraction of births that are low birth weight at birth (less than 2,500 grams), and the fraction of births that are very high birthweight at birth (more than 4,500 grams). These data are reported by state and year in *Vital Statistics* (U.S. DHHS). If changes in fertility affect the use of cesarean delivery only through changes in birth severity, then including these regressors should substantially reduce our coefficient of interest.

In fact, the inclusion of these state-level severity controls has little effect on our result. The coefficient on fertility is virtually unchanged. The results indicate that both

TABLE 3 **Specification Checks: Birth Severity and Birth Order**

Dependent Variable	(1) Cesarean	(2) Cesarean	(3) Fetal Distress	(4) Maternal Distress	(5) Cesarean
Log fertility	-.934 (.205)	-1.015 (.250)	-.695 (.274)	-.951 (.347)	-.883 (.211)
Neonatal mortality	-.018 (.012)	.015 (.015)	-.031 (.016)	.022 (.023)	-.019 (.014)
Low birthweight (<2,500 grams)	14.12 (4.671)	7.152 (5.713)	7.654 (6.213)	13.50 (8.388)	15.52 (4.706)
High birthweight (>4,200 grams)	12.10 (7.481)	18.65 (9.173)	.674 (9.691)	-3.131 (13.48)	12.32 (7.505)
Previous cesarean	5.761 (.064)	7.421 (.066)	-.483 (.060)	-.476 (.088)	5.761 (.064)
Breech presentation		2.951 (.032)			
Fetal distress		4.368 (.028)			
Maternal distress		2.001 (.037)			
Other complications		1.997 (.021)			
Fraction 1st births					.500 (.652)
Fraction 5th or more births					-.722 (1.570)

Notes: Standard errors are in parentheses. All regressions include the set of regressors presented in Table 2, along with a full set of state and year dummies. All regressions are run as logits. $N = 255,593$.

more low- and high-birthweight births lead to more cesarean deliveries, which is consistent with our priors, although there is a negative correlation between neonatal mortality and the cesarean rate.

An alternative approach to controlling for underlying riskiness is to include diagnosis indicators for the severity of the individual birth. We have not included diagnosis controls, other than an indicator for previous cesarean, in the previous analysis for two reasons. First, since they are coded by the ob/gyn, they are also potentially endogenous to the same economic factors determining the cesarean decision. If physicians are going to induce demand for a cesarean, they must indicate a diagnosis that justifies the use of this more intensive technique. Numerous studies have documented such "coding creep" within the context of Medicare's Prospective Payment System (PPS) (i.e., Carter, Newhouse, and Relles, 1990). Second, there was a major change in the diagnosis coding system in the middle of our sample period (1978). The time series for some of the severity indicators show marked changes between 1978 and 1979. Neither of these problems was important for the previous cesarean indicator.

Despite these problems, it is instructive to consider whether the inclusion of severity controls greatly affects our finding of interest. We therefore include in our basic regression framework four measures of a "problem birth": breech presentation, fetal distress, maternal distress, and other complications. In fact, as column (2) of Table 3 shows, our finding is somewhat strengthened when these controls are added. This further militates against an explanation for these results based on underlying riskiness. The severity indicators themselves are all highly significant. Conditional upon individual severity, the effects of state-level severity measures are weakened; the only significant remaining correlation is with high-birthweight births, and the coefficient on mortality is now positive.

In the next two columns of Table 3, we directly examine the relationship between two of these severity indicators and the fertility rate. There is a significant rise in the rate at which both fetal distress and maternal distress are diagnosed as fertility decreases. This could be construed as evidence that fertility affects the underlying riskiness of the birth. Alternatively, it could be evidence that financial incentives affected physician coding decisions; increased use of cesarean delivery must be justified by increased coded severity. This would be consistent with past evidence from the Medicare PPS. It is also consistent with the very rapid rise in documented birth severity over the 1970s, despite no obvious deterioration in this era in the health of pregnant women.

A second alternative hypothesis is that the fertility effect is operating through correlations between birth order and cesarean delivery. A number of studies note that cesareans are more likely for first births, as well as births that are the fifth or higher (Marieskind, 1979; Placek and Taffel, 1980; Tussing and Wojotowycz, 1992). Furthermore, as fertility declines, a larger fraction of births will be first births, although a smaller fraction will be of a higher order.²⁶ Thus, there is a potential for omitted variable bias from the exclusion of birth order (which is not reported in the NHDS) from our model.

We address this potential problem using state aggregate information on the distribution of births across birth order. In the final column of Table 3, we include, along with the state/year severity controls, state/year controls for the fraction of births of order 1, and the fraction of order 5 or more. Including these variables has little effect on the results. The variables themselves are insignificant. The result is similar if the individual-level severity controls are included as well.

²⁶ The effect on order 1 is much larger; if we sum the fraction of births that are of order 1 and order 5 or more, there is a strong negative correlation with fertility.

A third alternative is that our findings are not the result of a robust correlation, but rather are driven by the experience of one or two outlying states. Alternatively, our findings may simply reflect the experience of one outlying region where fertility was falling and taste for cesareans was rising. In results not reported, we have tried both excluding from our sample the five states with the largest fertility fall and the five states with the largest fertility rise over the 1970–1982 period, and allowing each of the four census regions to have its own specific time trend in fertility. In both cases, the correlation between fertility and cesarean delivery was robust.

Finally, this finding could be a function of changing insurance coverage of mothers. Unfortunately, we have been unable to control for source of payment in the regressions thus far because it is missing before 1977. As noted above, cesarean delivery is more likely for insured mothers, and Leibowitz (1990) shows that births are likely to be moved forward if insurance coverage becomes more generous. This evidence suggests that any bias from omitted insurance status measures is going against our finding of a negative correlation between fertility and cesarean delivery. Nevertheless, we can use the data on payer source for the period from 1977 onward to examine the effect of omitting insurance type from our regression.²⁷

In the first row of Table 4, we present our regression results for the 1977–1982 period only. The relationship of interest is robust to this sample period restriction; the coefficient is almost identical to that in Table 2 and is statistically significant. In the next row we include dummy variables for being privately insured or being uninsured. These variables have their expected signs, but the control for being uninsured is small and insignificant. There is little effect on the coefficient of interest, suggesting no bias from omitted insurance status variables.

The surprising finding of an equal rate for publicly insured (the excluded group) and uninsured patients may be a function of omitted characteristics correlated with payer status. A large literature documents the fact that the uninsured are in worse health along a number of dimensions (i.e., Kasper, 1986), which would bias them toward having more cesarean deliveries conditional on financial incentives. In row (3) of Table 4, we control for individual severity using the measures listed in Table 3. Doing so substantially reduces the coefficient on private pay and raises (in absolute value) the coefficient on uninsured, although the latter remains insignificant.

The data on payer source allow a further test of our hypothesis. If physicians are inducing cesarean delivery for financial purposes, then this response should be greatest among those with the most generous insurance coverage, and smallest for those with the least generous. That is, for uninsured patients who can be expected to pay only a small portion of their bill, there is little expected reward to performing a cesarean relative to a normal childbirth. For privately insured patients, where the fee differential is greatest, the rewards will be the greatest as well. Therefore, if financial incentives are driving the fertility/cesarean relationship, we should see that this relationship is strongest for those with private insurance and weakest for the uninsured.

We demonstrate that this is the case in the remaining rows of Table 4. We do so by augmenting our basic regression framework with interactions between payer status and the fertility rate. As we show in row (4), the private payer interaction is negative, indicating that fertility falls had their greatest impact on this population, and the uninsured interaction is positive, indicating the smallest impact of fertility changes on this group. That is, for the privately insured, the net effect of a 10% drop in fertility is a

²⁷ A related economic omitted variable is the labor force participation of the mothers in our sample. We do not have information on labor force attachment for our sample. As a proxy, we have tried in earlier work to control for the state unemployment rate and state average income. This had no effect on our relationship of interest.

TABLE 4 **Specification Checks: Payer Status**

	Log Fertility	Private Pay	Uninsured	Private Pay × Log Fertility	Uninsured × Log Fertility	Severity Controls?
(1)	-.901 (.430)					No
(2)	-.967 (.431)	.159 (.027)	-.016 (.034)			No
(3)	-.830 (.548)	.088 (.034)	-.057 (.043)			Yes
(4)	-.855 (.452)	-.839 (.719)	.966 (.956)	-.240 (.172)	.239 (.231)	No
(5)	-.758 (.440)	-1.190 (.590)		-.326 (.142)		No
(6)	-.989 (.432)		1.523 (.792)		.398 (.191)	No
(7)	-.574 (.574)	-1.893 (.891)	1.241 (1.187)	-.476 (.213)	.318 (.289)	Yes
(8)	-.463 (.560)	-2.249 (.736)		-.568 (.177)		Yes
(9)	-.914 (.549)		2.594 (.989)		.657 (.240)	Yes

Notes: Standard errors are in parentheses. All regressions include the set of regressors presented in Table 2, along with a full set of state and year dummies. All regressions are run as logits, for the 1977–1982 period only. Dependent variable is a cesarean section dummy. Severity controls are the four diagnosis indicators used in Table 3. $N = 114,917$.

rise in the cesarean rate of 1.15 percentage points; but for the uninsured, the net effect is a rise in the cesarean rate of only .65 percentage points. Neither interaction is significant, however, perhaps due to the collinearity in the insurance measures. In the next two rows, we therefore include these interactions separately, and they achieve significance in each case.

In the final three rows, we repeat this exercise including the severity controls. The story is similar; the findings are stronger, albeit less precisely estimated.²⁸ In fact, in this case we estimate that a 10% fall in fertility led to a rise in the cesarean rate for the privately insured of 1.1 percentage points but a rise for the uninsured of only .27 percentage points. Thus, overall, the results are supportive of our contention that the fertility coefficient measures a response to financial incentives: fertility falls induced cesarean increases in the very groups where the financial incentives were strongest.

6. Is there an asymmetric response to fertility changes?

■ In the initial interpretation of our results, we applied the coefficient on fertility to its 1970–1982 trend in order to assess the contribution of fertility declines to the growth of cesarean delivery. However, such an interpretation is appropriate only if the use of cesarean delivery reacts symmetrically to increases and decreases in the fertility rate. In fact, there is reason to suspect an asymmetric response of cesarean delivery to inducement incentives. Even if ob/gyns would like to switch back to normal delivery, it may be difficult to do so once cesarean delivery has become an accepted mode of

²⁸ The coefficients on the main effects (private pay and uninsured coefficients) are actually wrong signed in rows (4)–(9), but at the mean fertility rate the partial derivatives are correctly signed when the severity controls are included.

delivery for a given type of birth. The most likely explanation for such an asymmetry would be the evolution of malpractice rules, which use local standards of care in deciding whether treatment was appropriate. If cesarean delivery had evolved to be the standard of care for a type of birth, then this built-in irreversibility of malpractice standards would make it difficult to revert to vaginal childbirth.²⁹ More generally, a model with evolving practice "norms" could also yield a similar prediction of technological "stickiness."

If technological diffusion is "sticky," it has important implications for the evaluation of economic incentives; such hysteresis suggests potential long-run effects of short-run economic changes. It may also imply asymmetric responses to declines and increases in fertility, with the declining fertility of the early 1970s leading to differential adoption of cesarean delivery across areas, but the subsequent increase having no effect. Indeed, the time-series evidence would suggest that this is the case, as cesarean delivery continued to grow after 1976 despite a rise in the fertility rate. The time-series evidence, however, does not prove that there is an asymmetric response, because our analysis controls for general time effects in considering the effect of differential state fertility changes.

In this section, therefore, we test for an asymmetric response of the use of cesarean delivery to rising and falling fertility rates. We do so by creating for each state two subperiods: a "falling" period and a "nonfalling" period. Nationally, these periods would be represented by 1970–1975 and 1975–1982 respectively; for North Carolina, 1970–1976 and 1976–1982; and for Kansas, 1970–1973 and 1973–1982 (refer to Figures 2–4). One potentially problematic issue is how to treat the "turning point" year, in which the rapid decline ended and (for most states) a gradual rise began (i.e., 1976 for North Carolina). In the analysis below, we try three different approaches: including this year in the "falling" period, including it in the "nonfalling" period, and dropping it altogether.

We run regressions of the form

$$CSEC_{ij,t} = f(\alpha + \beta_1 X_{ij,t} + \beta_2 PREV + \beta_3 Z_{ij,t} + \beta_4 FALL_{j,t} + \beta_5 FERT_{j,t} + \beta_6 FERT_{j,t} \times FALL_{j,t} + \beta_7 \tau_t + \epsilon_{ij,t}), \quad (7)$$

where $FALL_{j,t}$ is a dummy that equals one if state j is in the "falling" sample in year t and zero otherwise. In this framework, β_6 measures the differential effect of fertility on cesarean delivery in the period when fertility is falling. If there is a symmetric response to falling and rising fertility, then β_5 should be equal to the previous estimates for the fertility effect, and β_6 should be equal to zero. There is some heterogeneity in the timing of the turning points across states: 21 states have their turning points in 1976, 7 in 1975, 3 in 1974, 17 in 1973, and 1 in 1972.

Table 5 presents the results of estimating this equation for the three different treatments of the turning-point year. In no case is there evidence of an asymmetric response. The response to fertility is equally strong in the period when fertility is rising and falling. This suggests that, at the margin, physicians are able to both increase and decrease intensity in response to financial incentives.

²⁹ Of course, if the irreversibility were symmetric, then practice patterns would never change. But there may be a systematic bias in evolving standards of care toward the newest standard, at the cost of possible reversibility to the old standard.

TABLE 5 **Asymmetric Response?**

Treatment of Turning-Point Year	(1) Include in Fall	(2) Include in Rise	(3) Drop
Log fertility	-.929 (.202)	-.874 (.207)	-.881 (.212)
Falling?	.234 (.580)	-.029 (.634)	-.005 (.650)
Fertility $\times$ falling	.055 (.137)	.002 (.150)	.003 (.153)
Number of observations	255,593	255,593	236,950

Notes: Standard errors are in parentheses. All regressions include the set of regressors presented in Table 2, along with a full set of state and year dummies. All regressions are run as logits. Dependent variable is a cesarean section dummy. Regression specification is equation (7) in text.

7. Conclusions

■ Understanding the agency relationship between physicians and patients (or their insurers) is critical to modelling the medical economy. While this relationship has been the source of numerous theoretical examinations over the past 30 years, there is still not a firm empirical basis for modelling the physician’s objective function. This article tries to contribute to that basis, by demonstrating that physician utilization of a new, more highly reimbursed technology was quite responsive to the financial environment facing ob/gyns. This finding suggests that physicians overused cesarean delivery relative to the level that would be chosen by a financially disinterested medical provider. The magnitude of this response, however, was fairly small, so financial incentives may not cause particularly large deviations from the noninducement equilibrium.

This finding may have more general implications for the optimal private and public response to the diffusion of new medical procedures. A number of observers have suggested that this diffusion is the primary driver behind rapidly rising medical costs in the United States (Newhouse, 1992; Weisbrod, 1991), and recent theoretical work (Goddeeris, 1984; Baumgardner, 1991) has even suggested that such technological advance in medicine may be welfare reducing. Cesarean delivery is fairly inexpensive relative to other “replacement” medical technologies. Yet its very rapid adoption despite a lack of convincing evidence that it leads to improvements in birth outcomes raises the possibility that this was not simply the case of a higher-quality technique replacing an inferior one. Our findings suggest that decoupling physician reimbursement levels from the cost of the technology that is used may help to restrain the diffusion of procedures whose additional benefit is exceeded by their incremental cost.

This work can be usefully extended in at least three directions. First, similar tests could be carried out for the adoption of other medical technologies in different time periods. This would be particularly useful for cases of very expensive replacement technologies with little convincing evidence of improved outcomes. Second, our article leaves largely unanswered the question of why cesarean delivery, which could be done fairly successfully from the start of the 20th century onward (Marieskind, 1979), spread so rapidly after 1970. Finally, there is a general sense in the medical literature that the level of cesarean delivery is “too high” (NIH, 1981). Such a conclusion should be drawn only after weighing more carefully the costs and benefits of this procedure.

While there is considerable evidence on the physiological and financial costs of cesarean delivery, there is little convincing evidence on its benefits. Assessing these benefits should be a high priority for future research.

Appendix

- Below we present the basic regression results when varying the specification of births/fertility.

TABLE A1 **Varying the Specification of Births/Fertility**
Dependent Variable Is a Dummy for Cesarean Delivery

	(1)	(2)	(3)	(4)
Log births	-.447 (.121)	-.908 (.208)		
Log population		.852 (.312)		
Log fertility (births/100 population)			-.923 (.201)	
Fertility				-.516 (.124)

Notes: Standard errors are in parentheses. All regressions include the set of regressors presented in Table 2, along with a full set of state and year dummies. All regressions are run as logits. $N = 255,593$.

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